

Jay Patel

Associate DevOps Engineer

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SUMMARY

Associate DevOps Engineer with nearly one year of hands-on production experience in AWS cloud infrastructure, IAM automation, and CI/CD pipelines. Proven ability to reduce manual toil through automation, improve deployment reliability, and deliver zero-downtime releases. Strong systems mindset grounded in an ICT engineering background, with a focus on reliability, security, and operational discipline.

EXPERIENCE

Associate DevOps Engineer

Feb 2025 – Present

Wonderlend Hubs

- Built a JIRA-integrated Python system that parses structured access requests, provisions IAM users, assigns groups, and updates ticket workflows automatically, reducing manual access-handling time by **~60%** and eliminating a recurring class of configuration errors across every onboarding cycle.
- Engineered a secure credential onboarding pipeline using AWS SES and Slack notifications with enforced first-login password-reset flows and MFA-ready account templates, standardising delivery for **10+ weekly onboarding requests** and improving audit trail completeness.
- Diagnosed and resolved CI/CD pipeline failures spanning IAM policy gaps, misconfigured environment variables, and container image issues across AWS-based microservices, contributing to **multiple successful production releases per week** with minimal service disruption.
- Root-cause-analysed recurring IAM and environment-related deployment blockers, documented fixes, and collaborated with senior engineers to apply cross-cutting remediations, measurably reducing mean-time-to-resolution for known failure patterns.

PROJECTS

ECS Blue-Green Deployment — Terraform Infrastructure *AWS ECS Fargate, Terraform, Python, GitLab CI, ALB*

- Designed a JSON-driven Terraform root module that dynamically discovers domains, ECS clusters, services, and ECR repositories via `fileset()` and `jsondecode()`, reducing per-service onboarding from multi-file Terraform edits to **a single JSON file**, provisioning ECR, cluster, service, and auto-scaling resources in one apply.
- Implemented a blue-green deployment layer with ALB listener rules and always-paired blue/green target groups controlled by a single `active_color` flag, enabling **zero-downtime releases** and instant rollback without service restarts by keeping both environments perpetually warm.
- Authored `sync_config.py` to propagate image tags from a central source-of-truth JSON to all service definitions before every Terraform run, and `detect_changes.py` with `child_generator.py` to generate GitLab child pipelines that **trigger builds only for changed services**, eliminating unnecessary rebuilds.
- Embedded CPU and memory target-tracking Application Auto Scaling policies inside the ECS module with per-service `min/max_capacity` configuration via JSON, and configured `lifecycle { ignore_changes }` to **preserve runtime autoscaling decisions** across repeated Terraform plan/apply cycles.

DocVault — Serverless Document Management Platform *AWS Lambda, S3, API Gateway, DynamoDB, IaC*

- Designed and deployed a static S3-hosted document management platform with API Gateway, Lambda, and DynamoDB for file and user management, automated end-to-end via a CI/CD pipeline using 100% IaC, processing **500+ file uploads daily** at **99.99% data availability** within AWS Free Tier constraints.

Resurge-Net — Risk-Aware Cloud Autoscaling

Python, CNN, LSTM/GRU, Time-Series Forecasting

- Built a proactive autoscaling framework using CNN + LSTM/GRU ensemble models trained on Alibaba Cluster Trace v2018, explicitly modelling workload uncertainty and rare spike events; designed tunable risk-based scaling policies that **reduced simulated SLA violations and total instance-hours** compared to reactive and mean-only baselines under identical capacity constraints, evaluated via MAE, RMSE, pinball loss, and cost-SLA tradeoff curves.

SKILLS

Cloud & Infrastructure	AWS (ECS Fargate, ECR, IAM, ALB, Lambda, S3, API Gateway, DynamoDB, SES, CloudWatch, AppAutoScaling), Terraform
CI/CD & Automation	GitLab CI/CD, Multi-project Child Pipelines, Docker, Python, Bash
Languages	Python, C, C++, SQL, Bash
Concepts	Blue-Green Deployments, Zero-Downtime Releases, Auto-scaling, IAM & Least Privilege, Microservices, Serverless, Cost Optimisation
Tooling	Linux, Git, JIRA, Slack Integrations

EDUCATION

BTech. in Information and Communication Technology	2022 – 2026
Pandit Deendayal Energy University, Gandhinagar, Gujarat	<i>CPI: 8.95 / 10 (after 7th sem)</i>

ACHIEVEMENTS

- Reduced manual AWS resource provisioning time by **>80%** through Terraform automation across 10+ resource types spanning ECR, ECS, ALB, and Application Auto Scaling.
- Delivered a 100% IaC serverless platform (DocVault) sustaining 500+ daily uploads at 99.99% availability within Free Tier constraints.
- Contributed to 5+ microservice deployment pipelines, improving build quality, rollback reliability, and overall pipeline success rates.
- AWS Certified Cloud Practitioner.